

When the rare outcome matters

Sampling, model comparison, thresholds, calibration, and subgroup checks at the ED triage boundary

The central question

Can triage-era information rank ED stays by the likelihood of a critical outcome?

Carry the data contract into modeling



Row grain

one row represents one ED stay



Timing

triage fields are separated from later events



Data quality

raw and derived imputed fields remain distinguishable

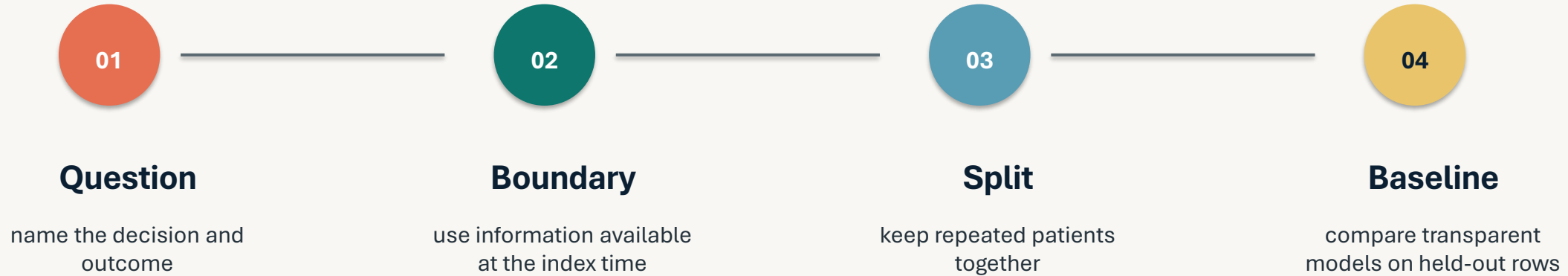


Leakage

identifiers, labels, and downstream fields stay out of X

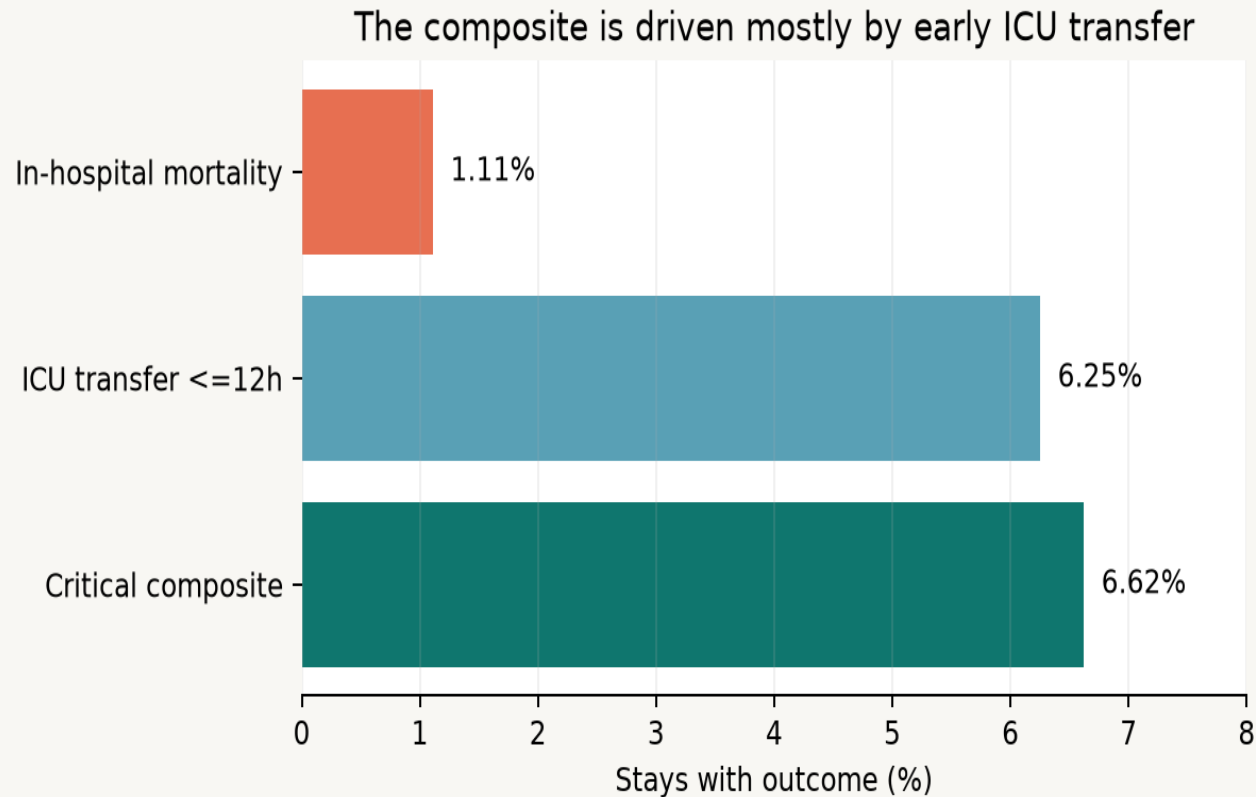
M3 inherits the row, timing, quality, and leakage decisions from M1; it changes the clinical question.

The modeling workflow already has a rhythm



M2 made the decision boundary explicit. M3 asks what changes when the positive class is rare.

Why this outcome matters



Severe events, one label

- Mortality and early ICU transfer both represent high-consequence outcomes.
- The composite can focus a triage-risk question on severe events.
- It also combines clinically different pathways, so components must remain visible.
- This is a retrospective benchmark, not a diagnosis or treatment recommendation.

The target is defined after the stay



Use triage-era inputs to estimate a later label. Never let the later label leak backward into X.

Rare positives change how metrics should be read

6.72%

positive prevalence in the held-out test set

401

positive test stays

5,563

negative test stays

0

patient overlap across train
and test

Why accuracy can mislead

- A majority-class dummy is accurate about 93% of the time.
- It finds zero critical cases.
- Sensitivity, PPV, average precision, and confusion counts make the trade-off visible.

The test population stays real

- Sampling changes training rows, not the held-out test population.
- PPV remains tied to the prevalence seen at evaluation.
- Patient-disjoint splitting tests generalization to unseen patients from this source.

Sampling changes what the learner sees

All five strategies use the same target, but they create different training distributions.



Original

keep the observed 93:7 class ratio



Under-sampling

remove some negatives to balance classes



Over-sampling

duplicate positive rows



SMOTE

create synthetic positive examples between neighbors

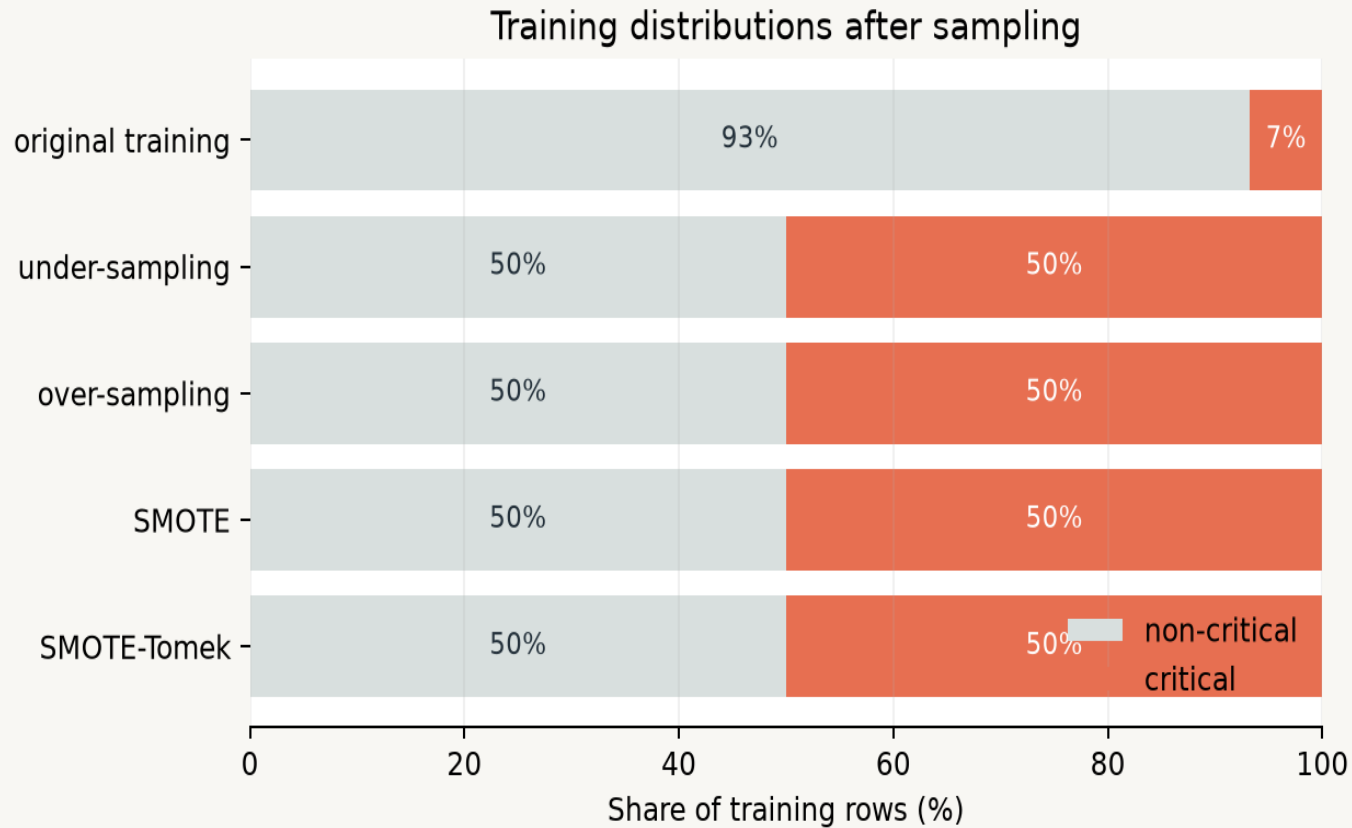


SMOTE-Tomek

SMOTE plus removal of ambiguous boundary pairs

Training balance is a learning aid, not a claim that the real ED population is balanced.

What the sampling experiment actually produced



Read the bars

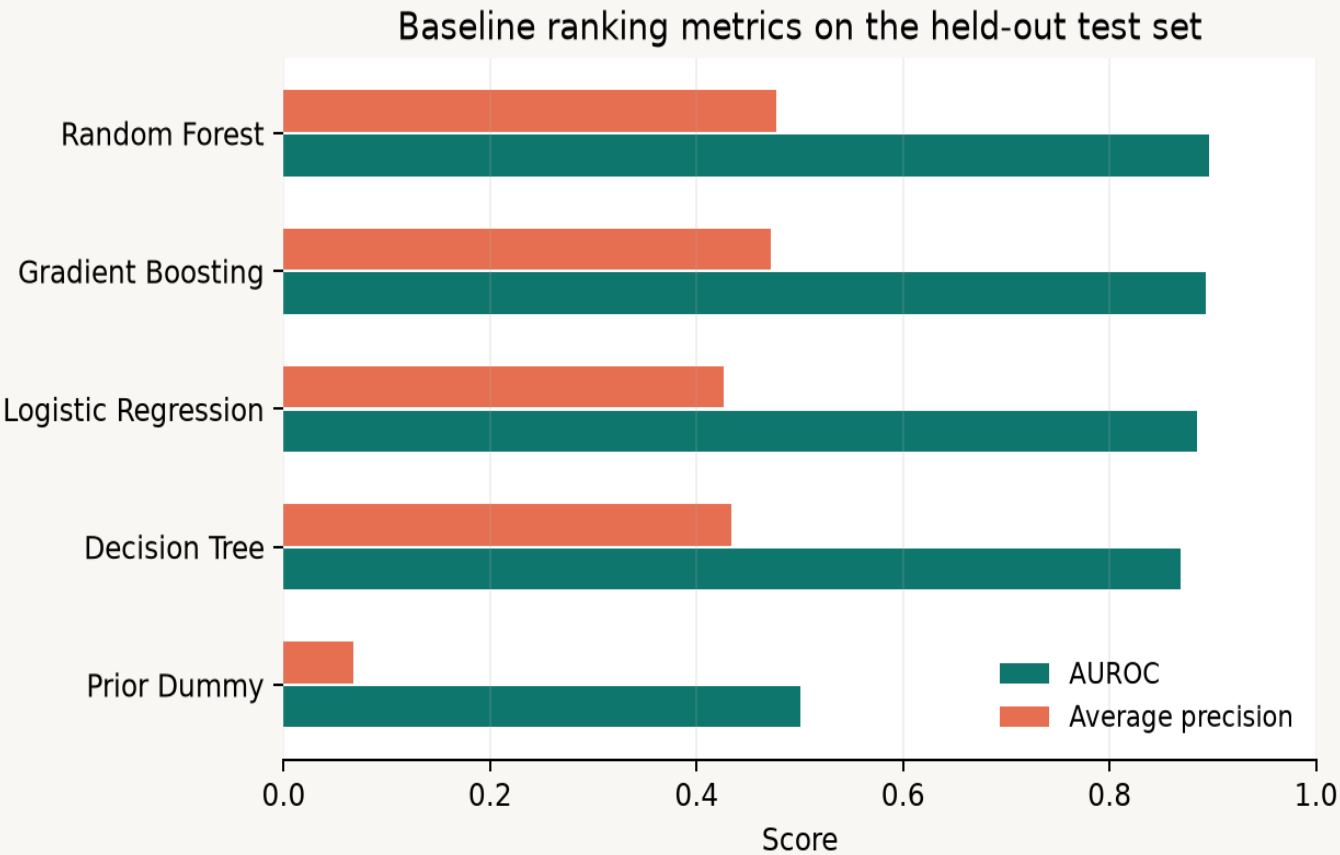
- Original training retains 22,408 negatives and 1,628 positives.
- Under-sampling keeps 1,628 rows per class.
- The other balanced methods create 50:50 training sets, with SMOTE-Tomek slightly trimming rows.
- The test set remains unbalanced and untouched.

Model families: same boundary, different assumptions

Prior dummy	uses prevalence only	reference floor
Logistic regression	linear signal after preprocessing	transparent baseline
Decision tree	rules and interactions	single-tree structure
Random forest	many trees averaged	nonlinear ensemble
Gradient boosting	trees correct earlier errors	sequential ensemble

A fair comparison holds the features, preprocessing, patient split, and test set constant.

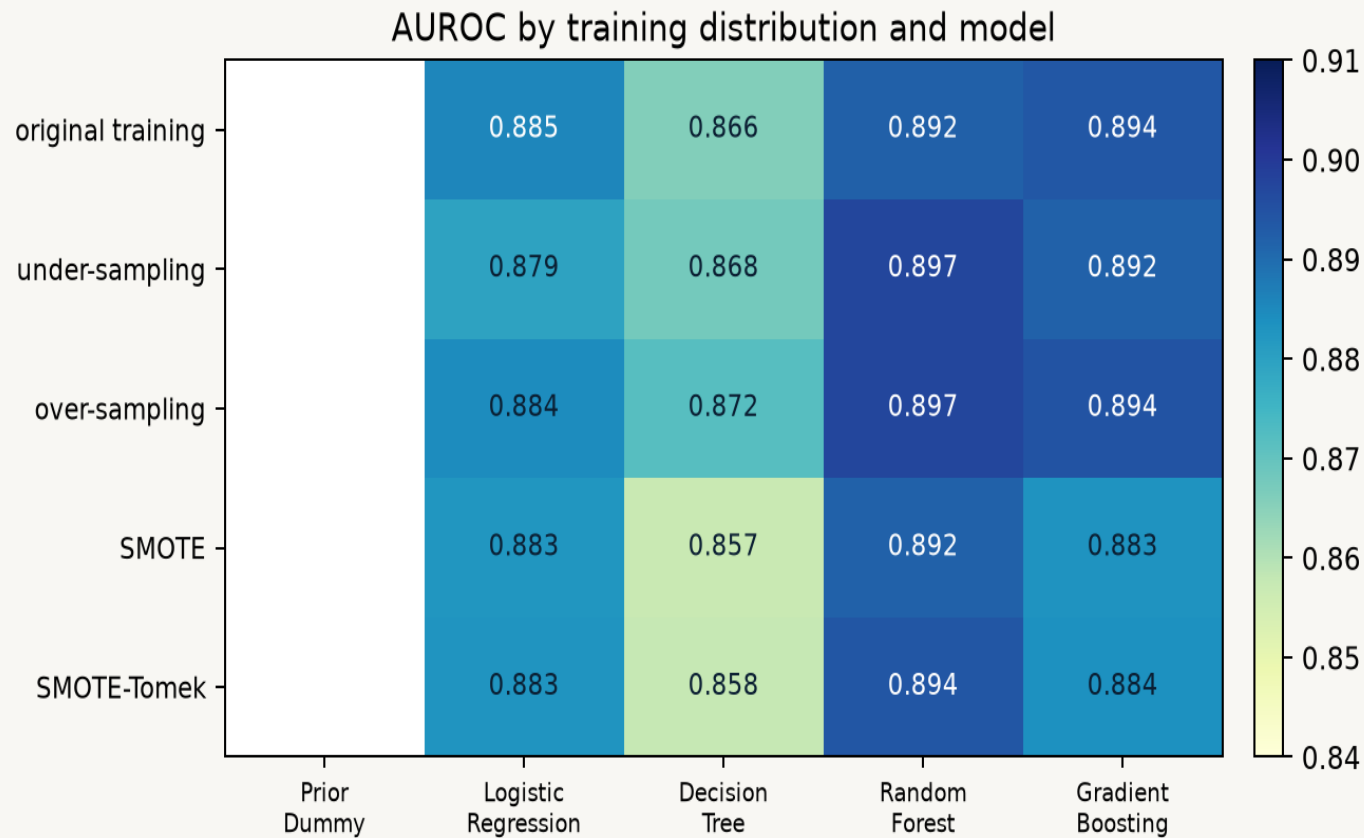
Baseline models: ranking is not the same as alert usefulness



Model	AUROC	Avg precision	Sensitivity	PPV
Random Forest	0.897	0.477	0.479	0.482
Gradient Boosting	0.894	0.471	0.172	0.734
Logistic Regression	0.885	0.426	0.823	0.234
Decision Tree	0.868	0.433	0.813	0.223
Prior Dummy	0.500	0.067	0.000	0.000

Best baseline AUROC: Random Forest at 0.897

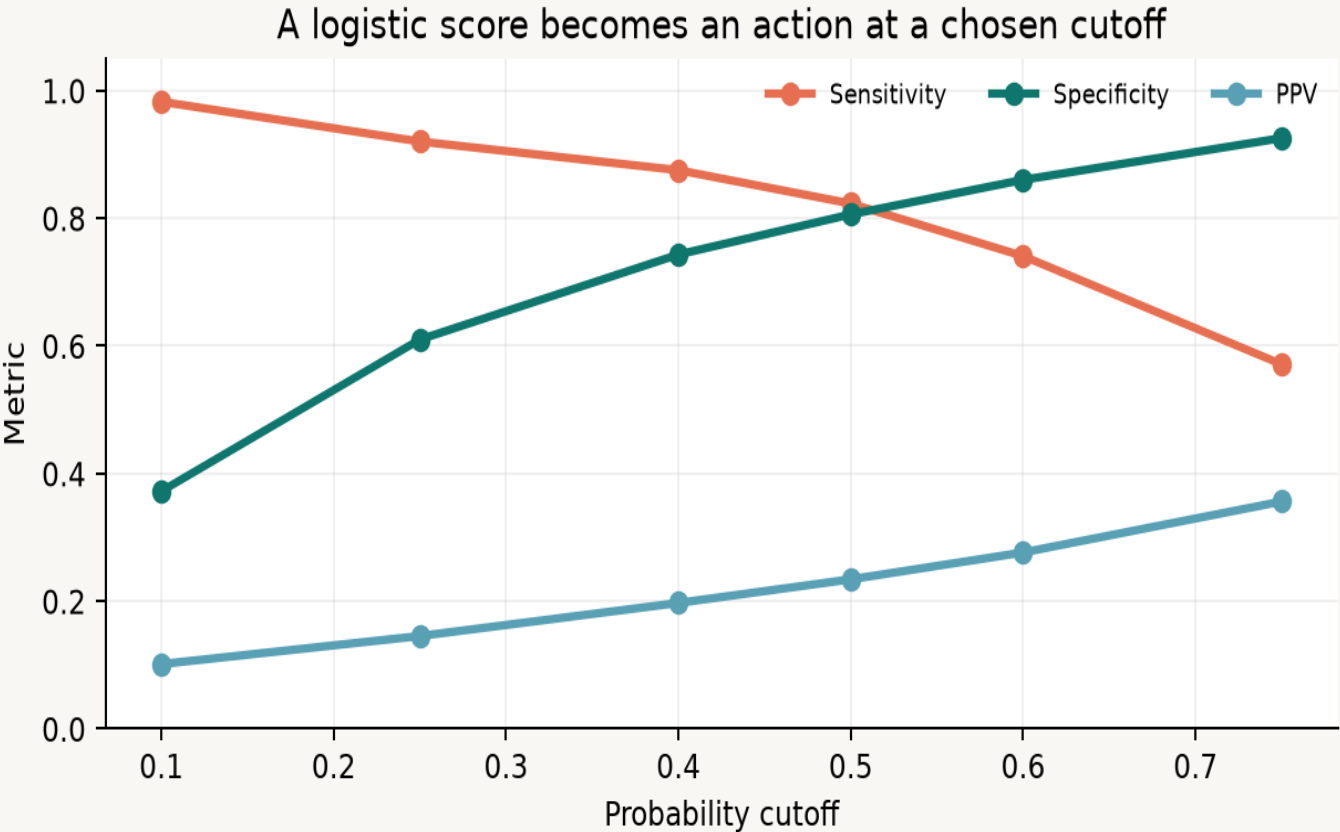
Sampling changes operating behavior more than ranking



Current reading

- Best sampled AUROC: Random Forest with random over-sampling (0.897).
- The ranking values remain in a fairly narrow band across sampling choices.
- Threshold metrics move more visibly: balancing can raise sensitivity while lowering specificity or PPV.
- Sampling should be selected with the intended action and calibration in mind, not AUROC alone.

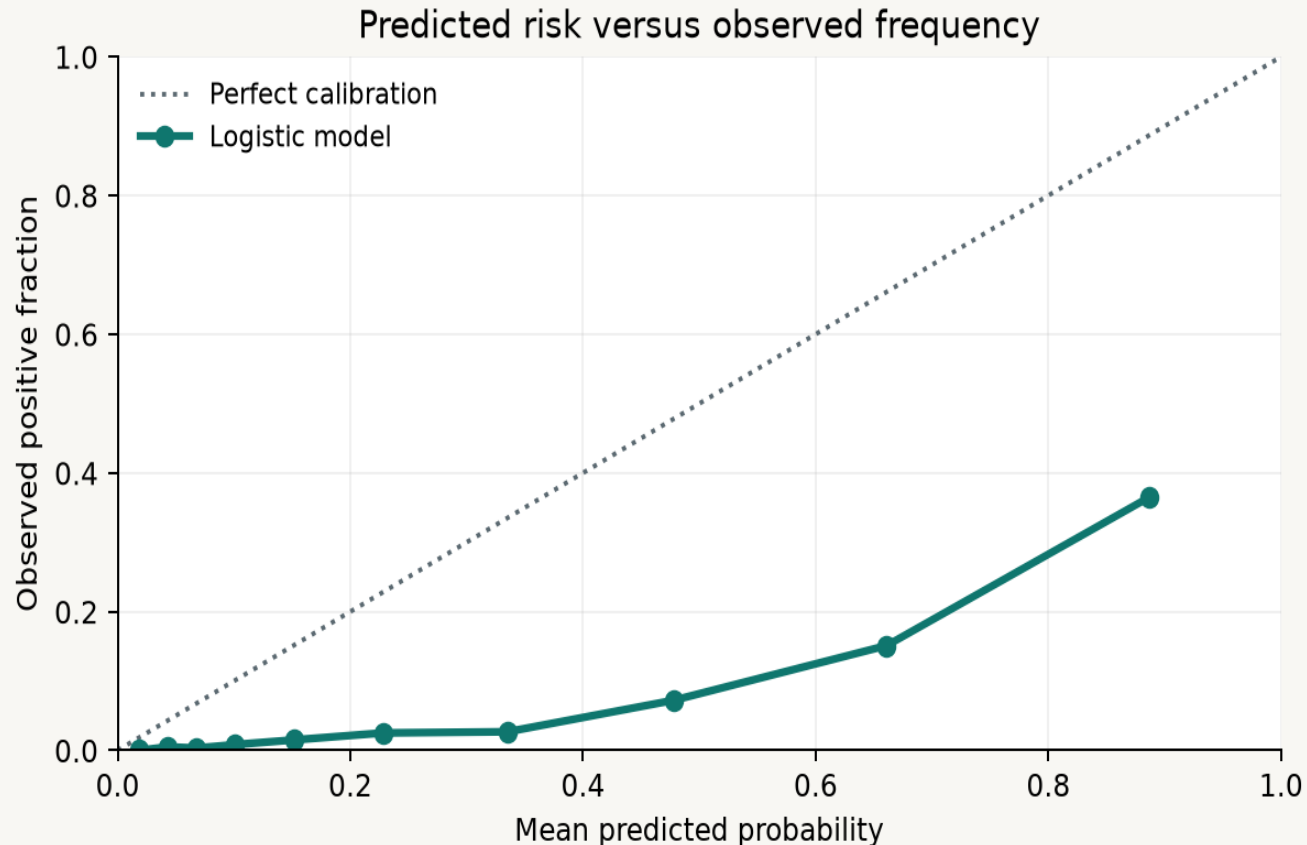
A probability becomes a decision at a threshold



Cutoff	Sensitivity	Specificity	PPV	False positives
0.10	0.983	0.372	0.101	3,492
0.25	0.920	0.610	0.145	2,170
0.50	0.823	0.806	0.234	1,079
0.75	0.571	0.926	0.356	414

Lower cutoff = more catches and more alerts. Higher cutoff = fewer alerts and more misses.

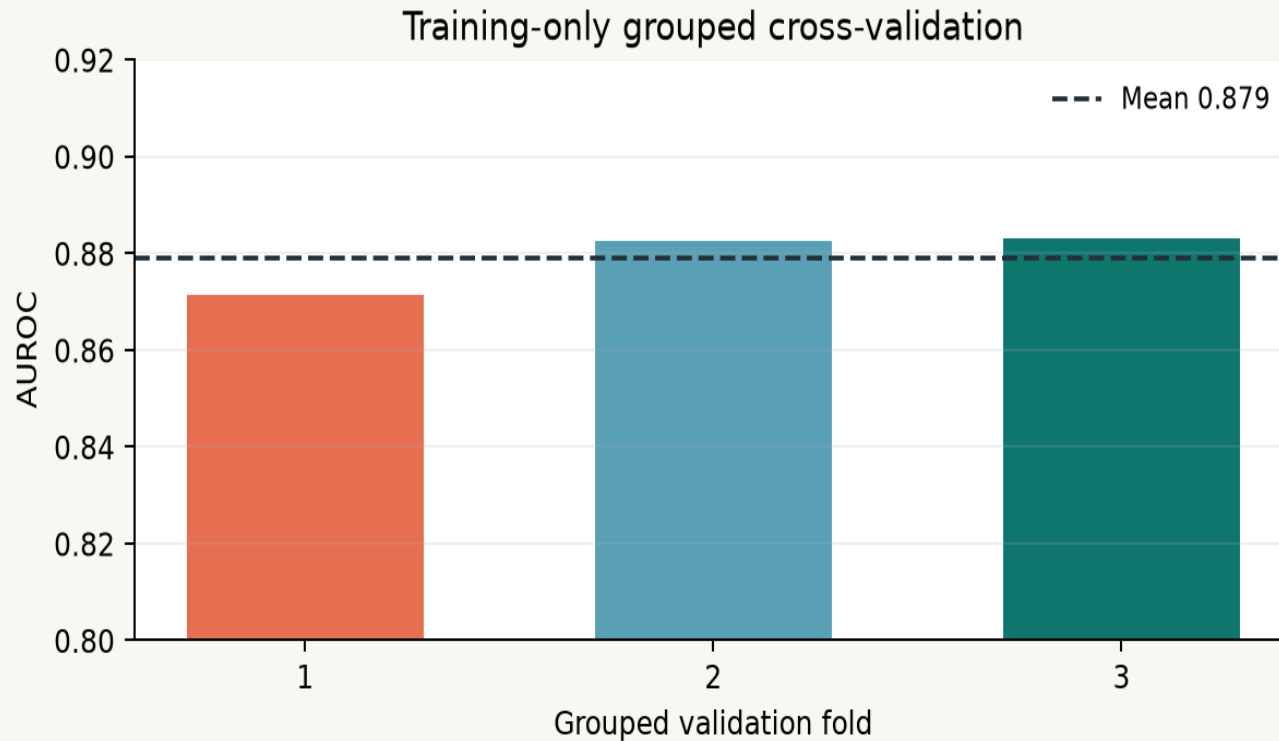
Calibration asks whether probabilities mean what they say



How to read the curve

- A calibrated 0.20 prediction means roughly 20% positive in similar cases.
- Points below the diagonal are overconfident: predicted risk exceeds observed frequency.
- The current curve is below the diagonal across much of the range.
- Sampling can shift probability meaning, so recalibration may be needed after resampling.

Cross-validation belongs inside training

**0.879**

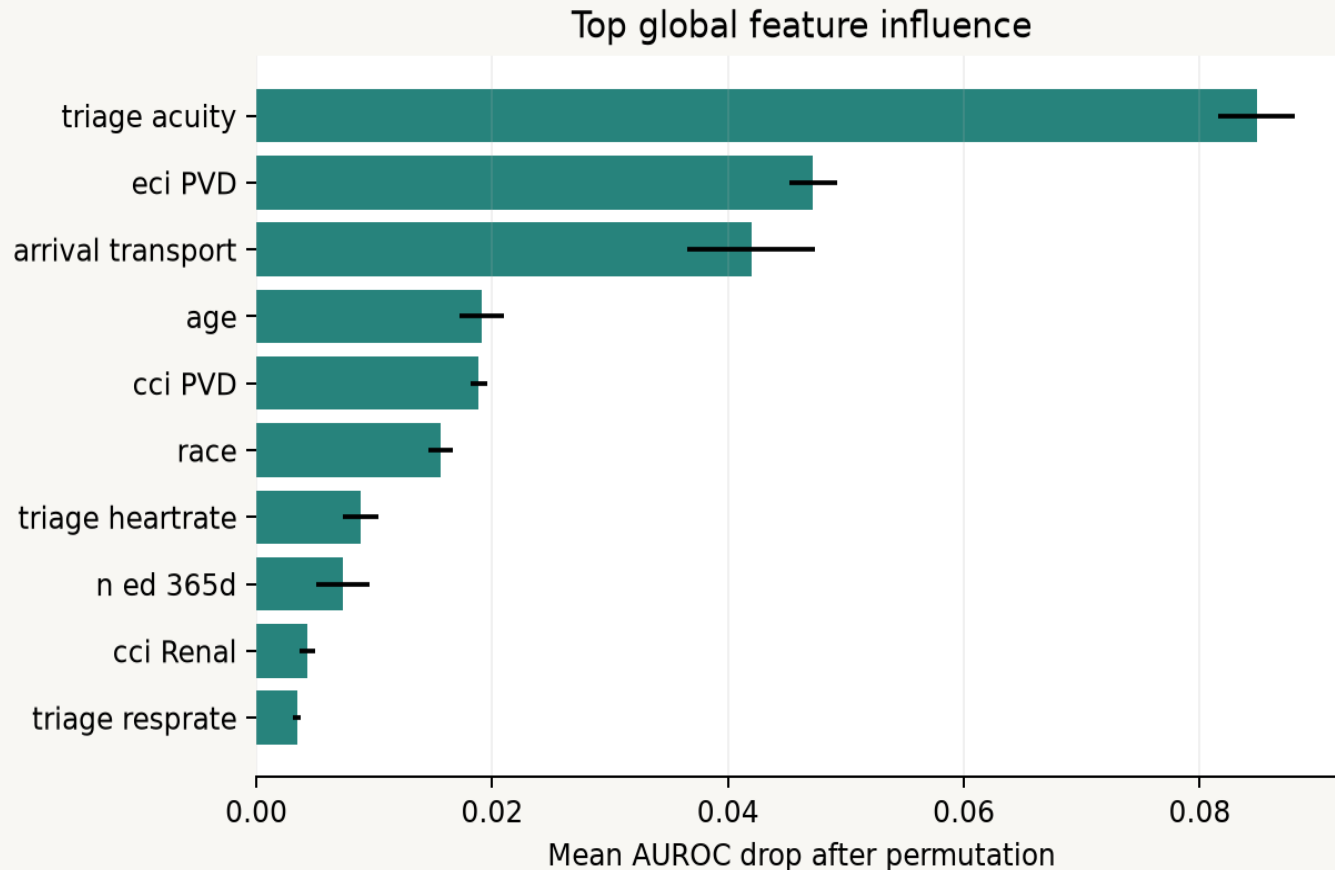
mean grouped-CV AUROC

+/- 0.007fold-to-fold standard
deviation

The boundary

- Grouped folds keep patients together.
- Fold variation is evidence about training stability.
- The held-out test set stays reserved for final comparison.
- External and temporal validation are still needed.

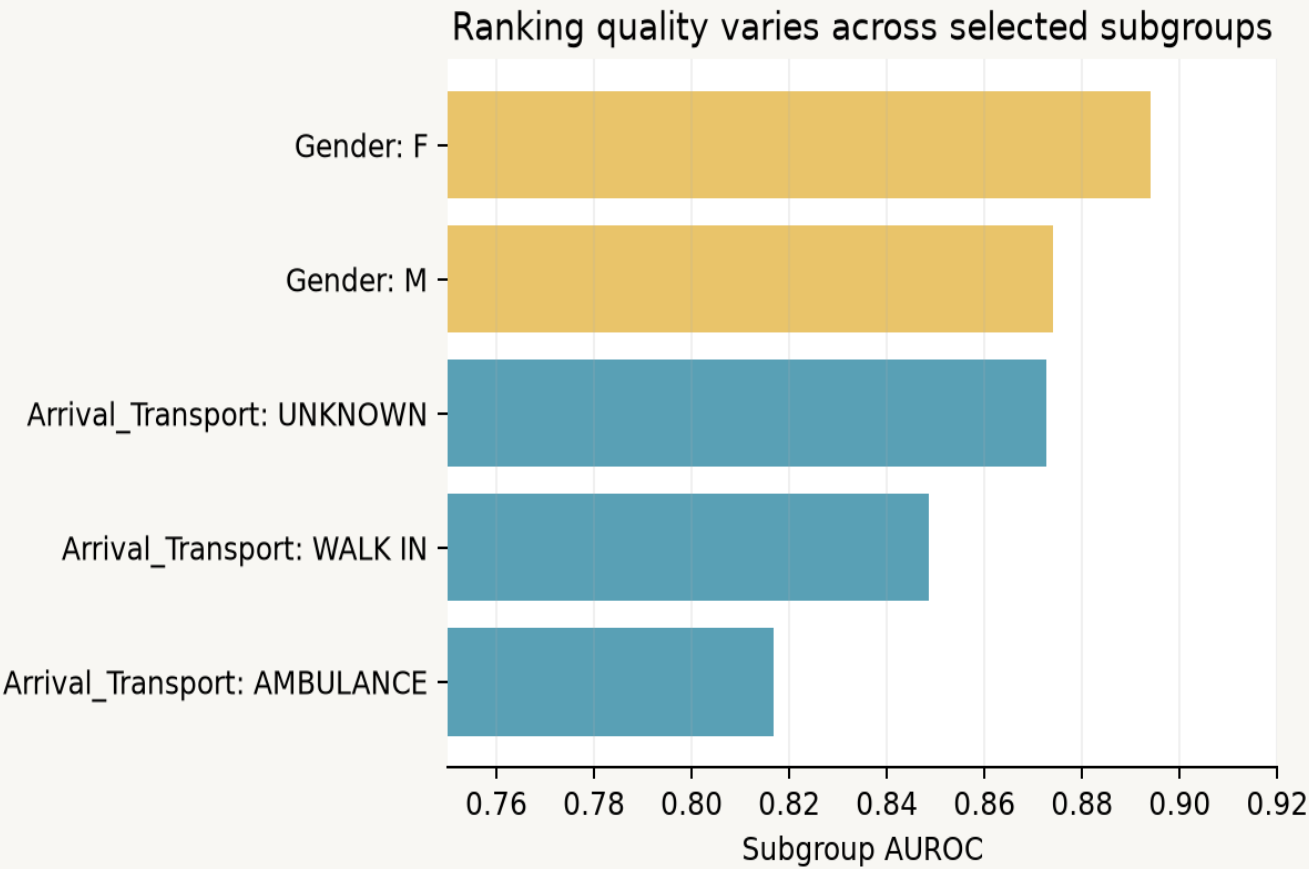
Global feature influence: useful, limited, and not causal



Permutation logic

- Shuffle one feature in held-out rows.
- Measure how much AUROC falls.
- A larger drop means the pipeline used that feature's information on these rows.
- Correlated features can share or mask one another's influence.
- Influence is not causality, clinical necessity, or fairness evidence.

Overall performance can hide uneven errors



Group	n	Prev.	Sens.	Spec.
gender: F	3,217	6.1%	0.811	0.830
gender: M	2,747	7.5%	0.834	0.778
arrival_transport: AMBULANCE	2,148	11.6%	0.880	0.571
arrival_transport: UNKNOWN	231	26.8%	1.000	0.503
arrival_transport: WALK IN	3,555	2.2%	0.468	0.950

Small groups need uncertainty, not confident ranking.

False positives and false negatives are the actual workflow

71

logistic false negatives at 0.50

1,079

logistic false positives at 0.50

1,150

exported example rows for inspection

False negative

A critical stay receives a low score and is not flagged at the chosen cutoff.

Question: what safety net catches the miss?

False positive

A non-critical stay receives a high score and consumes attention or capacity.

Question: what is the cost of the alert volume?

What this analysis can and cannot support

Retrospective labels

The composite is observed after triage and combines mortality with ICU transfer.

Single-source evidence

One extracted MIMIC-IV-ED teaching table does not establish transportability.

Model scope

This is a tabular baseline; free text, temporal drift, and clinical workflow are out of scope.

Uncertainty

Point estimates lack confidence intervals and small subgroups are unstable.

Operational validity

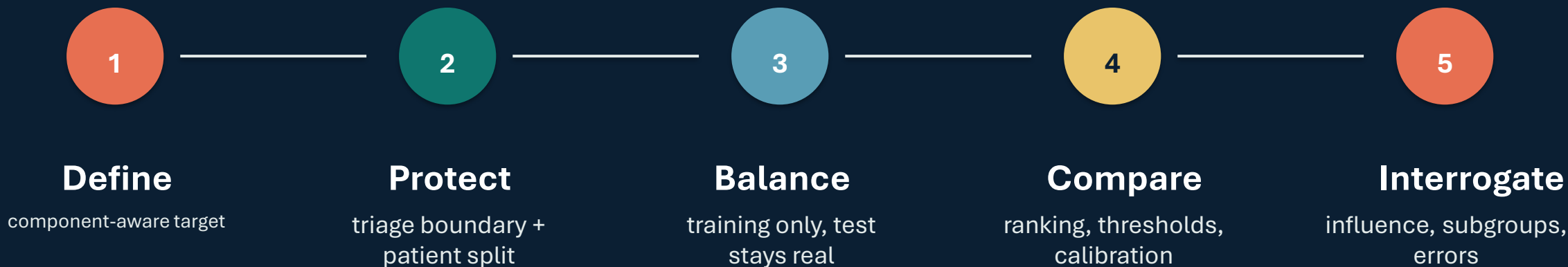
No prospective workflow, calibration intervention, or clinical utility study was performed.

Fairness

Subgroup metrics are an initial audit, not a complete fairness or harm assessment.

Responsible conclusion: these exports support learning and hypothesis generation, not deployment or clinical decisions.

The M3 mental model



A strong score is only one piece of a defensible medical ML argument.

Next: carry the same discipline into the hospitalization-specific target and leakage boundary.